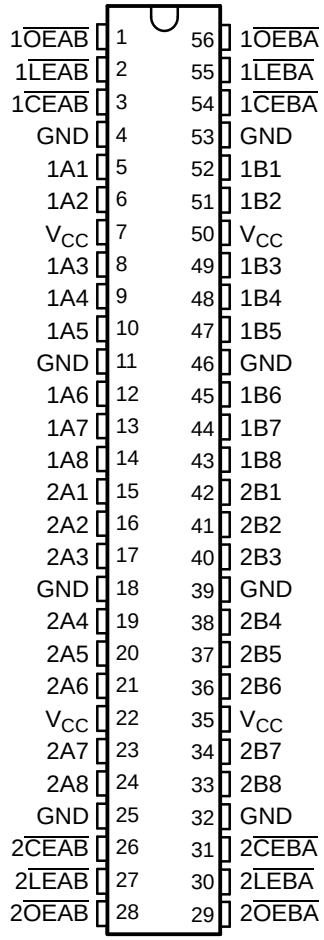
SN74LVCH16543A
16-BIT REGISTERED TRANSCEIVER WITH 3-STATE OUTPUTS

SCAS317M–NOVEMBER 1993–REVISED MARCH 2005

FEATURES

- **Member of the Texas Instruments Widebus™ Family**
- **Operates From 1.65 V to 3.6 V**
- **Inputs Accept Voltages to 5.5 V**
- **Max t_{pd} of 5.4 ns at 3.3 V**
- **Typical V_{OLP} (Output Ground Bounce) < 0.8 V at V_{CC} = 3.3 V, T_A = 25°C**
- **Typical V_{OHV} (Output V_{OH} Undershoot) > 2 V at V_{CC} = 3.3 V, T_A = 25°C**
- **I_{off} Supports Partial-Power-Down Mode Operation**
- **Supports Mixed-Mode Signal Operation on All Ports (5-V Input/Output Voltage With 3.3-V V_{CC})**
- **Bus Hold on Data Inputs Eliminates the Need for External Pullup/Pulldown Resistors**
- **Latch-Up Performance Exceeds 250 mA Per JESD 17**
- **ESD Protection Exceeds JESD 22**
  - **2000-V Human-Body Model (A114-A)**
  - **200-V Machine Model (A115-A)**
  - **1000-V Charged-Device Model (C101)**

DGG, DGV, OR DL PACKAGE
(TOP VIEW)



DESCRIPTION/ORDERING INFORMATION

This 16-bit registered transceiver is designed for 1.65-V to 3.6-V V_{CC} operation.

The SN74LVCH16543A can be used as two 8-bit transceivers or one 16-bit transceiver. Separate latch-enable ($\overline{LEAB}$ or $\overline{LEBA}$) and output-enable ($\overline{OEAB}$ or $\overline{OEBA}$) inputs are provided for each register, to permit independent control in either direction of data flow.

ORDERING INFORMATION

T_A	PACKAGE(1)		ORDERABLE PART NUMBER	TOP-SIDE MARKING
–40°C to 85°C	SSOP – DL	Tube	SN74LVCH16543ADL	LVCH16543A
		Tape and reel	SN74LVCH16543ADLR	
	TSSOP – DGG	Tape and reel	SN74LVCH16543ADGGR	LVCH16543A
	TVSOP – DGV	Tape and reel	SN74LVCH16543ADGVR	LDH543A
	VFBGA – GQL	Tape and reel	SN74LVCH16543AGQLR	LDH543A
	VFBGA – ZQL (Pb-free)		SN74LVCH16543AZQLR	

(1) Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.

Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

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PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of the Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

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DESCRIPTION/ORDERING INFORMATION (CONTINUED)

The A-to-B enable ($\overline{CEAB}$) input must be low to enter data from A or to output data from B. If $\overline{CEAB}$ is low and $\overline{LEAB}$ is low, the A-to-B latches are transparent; a subsequent low-to-high transition of $\overline{LEAB}$ puts the A latches in the storage mode. With $\overline{CEAB}$ and $\overline{OEAB}$ both low, the 3-state B outputs are active and reflect the data present at the output of the A latches. Data flow from B to A is similar, but requires using the $\overline{CEBA}$, $\overline{LEBA}$, and $\overline{OEBA}$ inputs.

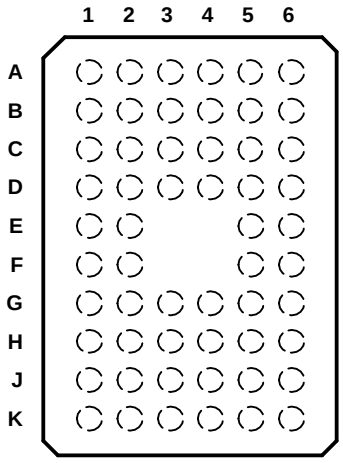
To ensure the high-impedance state during power up or power down, $\overline{OE}$ should be tied to V_{CC} through a pullup resistor; the minimum value of the resistor is determined by the current-sinking capability of the driver.

Inputs can be driven from either 3.3-V or 5-V devices. This feature allows the use of this device as a translator in a mixed 3.3-V/5-V system environment.

This device is fully specified for partial-power-down applications using I_{off}. The I_{off} circuitry disables the outputs, preventing damaging current backflow through the device when it is powered down.

Active bus-hold circuitry is provided to hold unused or floating data inputs at a valid logic level. Use of pullup or pulldown resistors with the bus-hold circuitry is not recommended. The bus-hold circuitry is part of the input circuit and is not disabled by $\overline{OE}$ or DIR.

GQL OR ZQL PACKAGE
(TOP VIEW)



TERMINAL ASSIGNMENTS

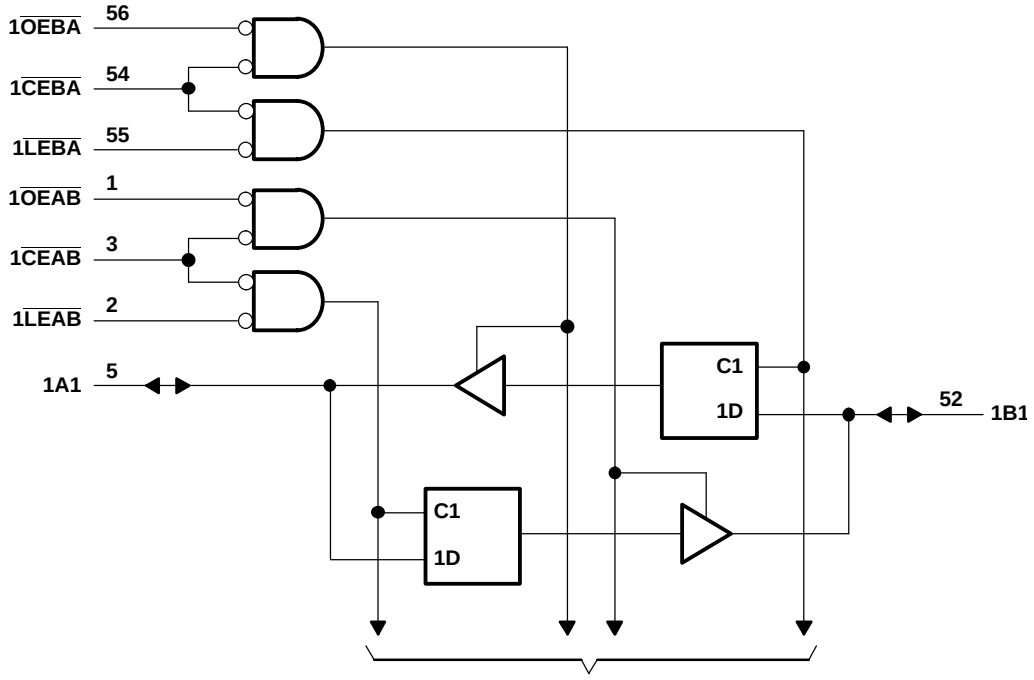
	1	2	3	4	5	6
A	1$\overline{CEAB}$	1$\overline{LEAB}$	1$\overline{OEAB}$	1$\overline{OEBA}$	1$\overline{LEBA}$	1$\overline{CEBA}$
B	1A2	1A1	GND	GND	1B1	1B2
C	1A4	1A3	V_{CC}	V_{CC}	1B3	1B4
D	1A6	1A5	GND	GND	1B5	1B6
E	1A8	1A7			1B7	1B8
F	2A1	2A2			2B2	2B1
G	2A3	2A4	GND	GND	2B4	2B3
H	2A5	2A6	V_{CC}	V_{CC}	2B6	2B5
J	2A7	2A8	GND	GND	2B8	2B7
K	2$\overline{CEAB}$	2$\overline{LEAB}$	2$\overline{OEAB}$	2$\overline{OEBA}$	2$\overline{LEBA}$	2$\overline{CEBA}$

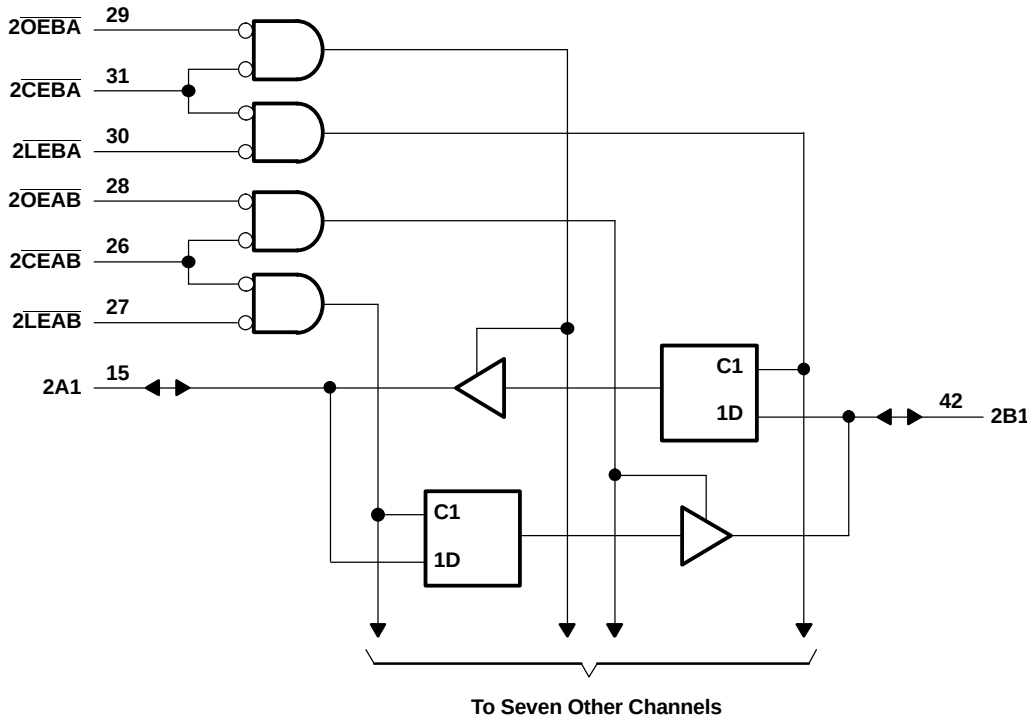
FUNCTION TABLE(1)
(EACH 8-BIT SECTION)

INPUTS				OUTPUT
$\overline{CEAB}$	$\overline{LEAB}$	$\overline{OEAB}$	A	B
H	X	X	X	Z
X	X	H	X	Z
L	H	L	X	B_0(2)
L	L	L	L	L
L	L	L	H	H

(1) A-to-B data flow is shown; B-to-A flow control is the same, except that it uses $\overline{CEBA}$, $\overline{LEBA}$, and $\overline{OEBA}$.
(2) Output level before the indicated steady-state input conditions were established

LOGIC DIAGRAM (POSITIVE LOGIC)





Pin numbers shown are for the DGG, DGV, and DL packages.

Absolute Maximum Ratings(1)

over operating free-air temperature range (unless otherwise noted)

			MIN	MAX	UNIT
V_{CC}	Supply voltage range		–0.5	6.5	V
V_I	Input voltage range(2)		–0.5	6.5	V
V_O	Voltage range applied to any output in the high-impedance or power-off state(2)		–0.5	6.5	V
V_O	Voltage range applied to any output in the high or low state(2)(3)		–0.5	V_{CC} + 0.5	V
I_{IK}	Input clamp current	V_I < 0		–50	mA
I_{OK}	Output clamp current	V_O < 0		–50	mA
I_O	Continuous output current			±50	mA
	Continuous current through each V_{CC} or GND			±100	mA
θ_{JA}	Package thermal impedance(4)	DGG package		64	°C/W
		DGV package		48	
		DL package		56	
		GQL/ZQL package		42	
T_{stg}	Storage temperature range		–65	150	°C

(1) Stresses beyond those listed under "absolute maximum ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under "recommended operating conditions" is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) The input negative-voltage and output voltage ratings may be exceeded if the input and output current ratings are observed.
(3) The value of V_{CC} is provided in the recommended operating conditions table.
(4) The package thermal impedance is calculated in accordance with JESD 51-7.

Recommended Operating Conditions(1)

			MIN	MAX	UNIT
V_{CC}	Supply voltage	Operating	1.65	3.6	V
		Data retention only	1.5		
V_{IH}	High-level input voltage	V_{CC} = 1.65 V to 1.95 V	0.65 × V_{CC}		V
		V_{CC} = 2.3 V to 2.7 V	1.7		
		V_{CC} = 2.7 V to 3.6 V	2		
V_{IL}	Low-level input voltage	V_{CC} = 1.65 V to 1.95 V		0.35 × V_{CC}	V
		V_{CC} = 2.3 V to 2.7 V		0.7	
		V_{CC} = 2.7 V to 3.6 V		0.8	
V_I	Input voltage		0	5.5	V
V_O	Output voltage	High or low state	0	V_{CC}	V
		3-state	0	5.5	
I_{OH}	High-level output current	V_{CC} = 1.65 V		–4	mA
		V_{CC} = 2.3 V		–8	
		V_{CC} = 2.7 V		–12	
		V_{CC} = 3 V		–24	
I_{OL}	Low-level output current	V_{CC} = 1.65 V		4	mA
		V_{CC} = 2.3 V		8	
		V_{CC} = 2.7 V		12	
		V_{CC} = 3 V		24	
$\Delta t/\Delta v$	Input transition rise or fall rate			10	ns/V
T_A	Operating free-air temperature		–40	85	°C

(1) All unused control inputs of the device must be held at V_{CC} or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.

Electrical Characteristics

over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS	V_{CC}	MIN	TYP(1)	MAX	UNIT
V_{OH}		I_{OH} = –100 µA	1.65 V to 3.6 V	V_{CC} – 0.2			V
		I_{OH} = –4 mA	1.65 V	1.2			
		I_{OH} = –8 mA	2.3 V	1.7			
		I_{OH} = –12 mA	2.7 V	2.2			
			3 V	2.4			
		I_{OH} = –24 mA	3 V	2.2			
V_{OL}		I_{OL} = 100 µA	1.65 V to 3.6 V			0.2	V
		I_{OL} = 4 mA	1.65 V			0.45	
		I_{OL} = 8 mA	2.3 V			0.7	
		I_{OL} = 12 mA	2.7 V			0.4	
		I_{OL} = 24 mA	3 V			0.55	
I_I	Control inputs	V_I = 0 to 5.5 V	3.6 V			±5	µA
I_{off}		V_I or V_O = 5.5 V	0			±10	µA
$I_{I(hold)}$	A or B ports	V_I = 0.58 V	1.65 V	(2)			µA
		V_I = 1.07 V		(2)			
		V_I = 0.7 V	2.3 V	45			
		V_I = 1.7 V		–45			
		V_I = 0.8 V	3 V	75			
		V_I = 2 V		–75			
		V_I = 0 to 3.6 V(3)	3.6 V			±500	
I_{OZ}(4)		V_O = 0 V or (V_{CC} to 5.5 V)	2.3 V to 3.6 V			±5	µA
I_{CC}		V_I = V_{CC} or GND, I_O = 0	3.6 V			20	µA
		3.6 V ≤ V_I ≤ 5.5 V(5), I_O = 0				20	
ΔI_{CC}		One input at V_{CC} – 0.6 V, Other inputs at V_{CC} or GND	2.7 V to 3.6 V			500	µA
C_i	Control inputs	V_I = V_{CC} or GND	3.3 V		5		pF
C_{io}	A or B ports	V_O = V_{CC} or GND	3.3 V		8		pF

(1) All typical values are at V_{CC} = 3.3 V, T_A = 25°C.
(2) This information was not available at the time of publication.
(3) This is the bus-hold maximum dynamic current required to switch the input from one state to another.
(4) For the total leakage current in an I/O port, consult the $I_{I(hold)}$ specification for the input voltage condition, 0 V < V_I < V_{CC}, and the I_{OZ} specification for the input voltage conditions, V_I = 0 V or V_I = V_{CC} to 5.5 V. The bus-hold current, at input voltage greater than V_{CC}, is negligible.
(5) This applies in the disabled state only.

Timing Requirements

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 1)

		V_{CC} = 1.8 V ± 0.15 V		V_{CC} = 2.5 V ± 0.2 V		V_{CC} = 2.7 V		V_{CC} = 3.3 V ± 0.3 V		UNIT
		MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	
t_w	Pulse duration, $\overline{LE}$ or $\overline{CE}$ low	(1)		(1)		3.3		3.3		ns
t_{su}	Setup time, data before $\overline{LE}$ or $\overline{CE}$↓	(1)		(1)		1.1		1.1		ns
t_h	Hold time, data after $\overline{LE}$ or $\overline{CE}$↓	(1)		(1)		1.9		1.9		ns

(1) This information was not available at the time of publication.

Switching Characteristics

over recommended operating free-air temperature range (unless otherwise noted) (see Figure 1)

PARAMETER	FROM (INPUT)	TO (OUTPUT)	V_{CC} = 1.8 V ± 0.15 V		V_{CC} = 2.5 V ± 0.2 V		V_{CC} = 2.7 V		V_{CC} = 3.3 V ± 0.3 V		UNIT
			MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	
t_{pd}	A or B	B or A	(1)	(1)	(1)	(1)		6.1	1.2	5.4	ns
	$\overline{LE}$	A or B	(1)	(1)	(1)	(1)		7.4	1.5	6.1	
t_{en}	$\overline{CE}$	A or B	(1)	(1)	(1)	(1)		7.9	1.2	6.6	ns
t_{dis}			(1)	(1)	(1)	(1)		7.1	1.5	6.6	
t_{en}	$\overline{OE}$	A or B	(1)	(1)	(1)	(1)		7.6	1	6.3	ns
t_{dis}			(1)	(1)	(1)	(1)		6.9	1.5	6.3	

(1) This information was not available at the time of publication.

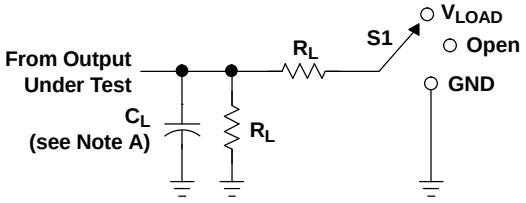
Operating Characteristics

T_A = 25°C

PARAMETER			TEST CONDITIONS	V_{CC} = 1.8 V	V_{CC} = 2.5 V	V_{CC} = 3.3 V	UNIT
				TYP	TYP	TYP	
C_{pd}	Power dissipation capacitance per transceiver	Outputs enabled	f = 10 MHz	(1)	(1)	44	pF
		Outputs disabled		(1)	(1)	4	

(1) This information was not available at the time of publication.

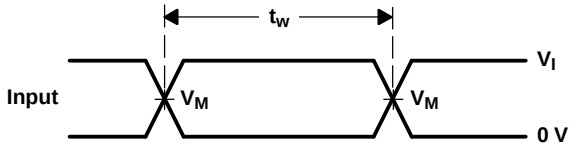
PARAMETER MEASUREMENT INFORMATION



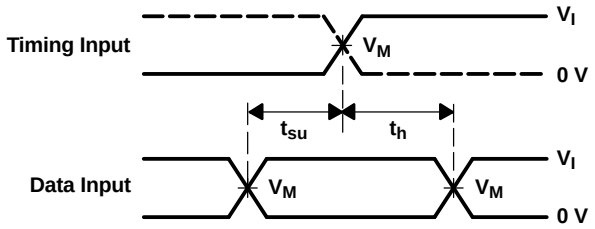
LOAD CIRCUIT

TEST	S1
t_{PLH}/t_{PHL}	Open
t_{PLZ}/t_{PZL}	V_{LOAD}
t_{PHZ}/t_{PZH}	GND

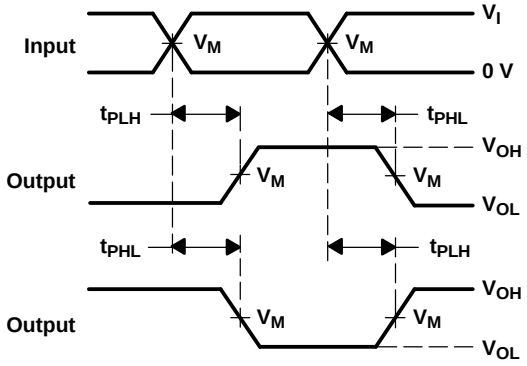
V_{CC}	INPUTS		V_M	V_{LOAD}	C_L	R_L	V_Δ
	V_I	t_r/t_f					
1.8 V ± 0.15 V	V_{CC}	≤2 ns	$V_{CC}/2$	$2 \times V_{CC}$	30 pF	1 kΩ	0.15 V
2.5 V ± 0.2 V	V_{CC}	≤2 ns	$V_{CC}/2$	$2 \times V_{CC}$	30 pF	500 Ω	0.15 V
2.7 V	2.7 V	≤2.5 ns	1.5 V	6 V	50 pF	500 Ω	0.3 V
3.3 V ± 0.3 V	2.7 V	≤2.5 ns	1.5 V	6 V	50 pF	500 Ω	0.3 V



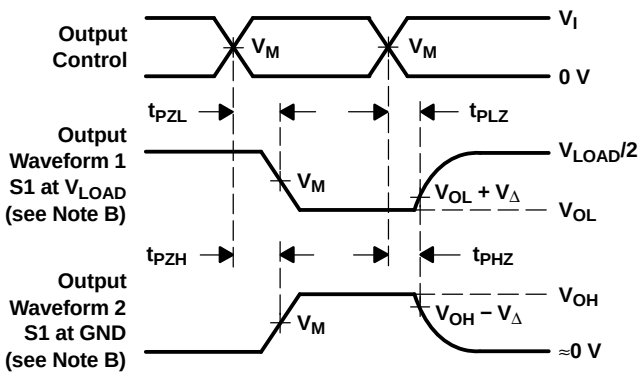
VOLTAGE WAVEFORMS
PULSE DURATION



VOLTAGE WAVEFORMS
SETUP AND HOLD TIMES



VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES
INVERTING AND NONINVERTING OUTPUTS



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES
LOW- AND HIGH-LEVEL ENABLING

NOTES: A. C_L includes probe and jig capacitance.
B. Waveform 1 is for an output with internal conditions such that the output is low, except when disabled by the output control. Waveform 2 is for an output with internal conditions such that the output is high, except when disabled by the output control.
C. All input pulses are supplied by generators having the following characteristics: PRR ≤ 10 MHz, Z_O = 50 Ω.
D. The outputs are measured one at a time, with one transition per measurement.
E. t_{PLZ} and t_{PHZ} are the same as t_{dis}.
F. t_{PZL} and t_{PZH} are the same as t_{en}.
G. t_{PLH} and t_{PHL} are the same as t_{pd}.
H. All parameters and waveforms are not applicable to all devices.

Figure 1. Load Circuit and Voltage Waveforms

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
74LVCH16543ADLRG4	Active	Production	SSOP (DL) | 56	1000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	LVCH16543A
74LVCH16543ADLRG4.B	Active	Production	SSOP (DL) | 56	1000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	LVCH16543A
SN74LVCH16543ADGGR	Active	Production	TSSOP (DGG) | 56	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	LVCH16543A
SN74LVCH16543ADGGR.B	Active	Production	TSSOP (DGG) | 56	2000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	LVCH16543A
SN74LVCH16543ADL	Active	Production	SSOP (DL) | 56	20 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	LVCH16543A
SN74LVCH16543ADL.B	Active	Production	SSOP (DL) | 56	20 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	LVCH16543A
SN74LVCH16543ADLR	Active	Production	SSOP (DL) | 56	1000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	LVCH16543A
SN74LVCH16543ADLR.B	Active	Production	SSOP (DL) | 56	1000 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	LVCH16543A

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

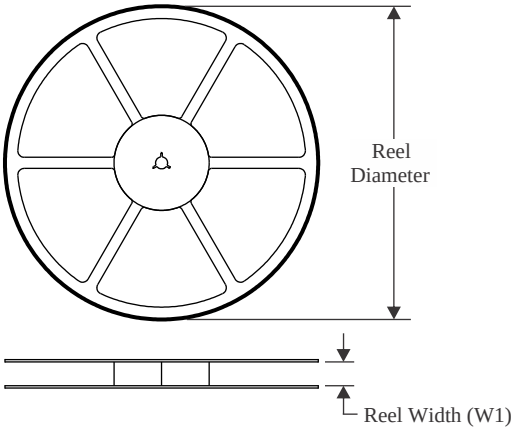
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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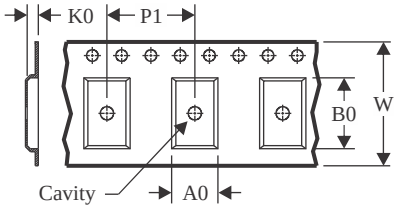
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TAPE AND REEL INFORMATION

REEL DIMENSIONS

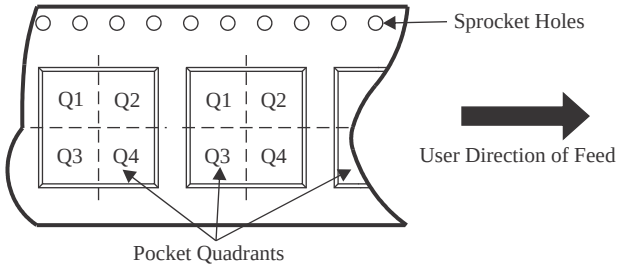


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

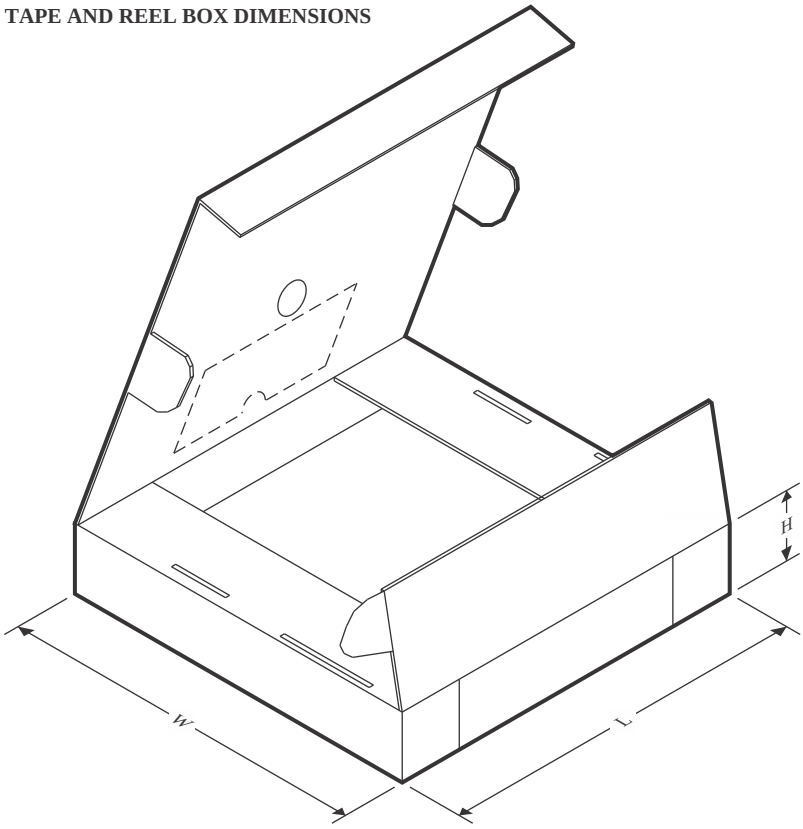
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
74LVCH16543ADLRG4	SSOP	DL	56	1000	330.0	32.4	11.35	18.67	3.1	16.0	32.0	Q1
SN74LVCH16543ADGGR	TSSOP	DGG	56	2000	330.0	24.4	8.9	14.7	1.4	12.0	24.0	Q1
SN74LVCH16543ADLR	SSOP	DL	56	1000	330.0	32.4	11.35	18.67	3.1	16.0	32.0	Q1

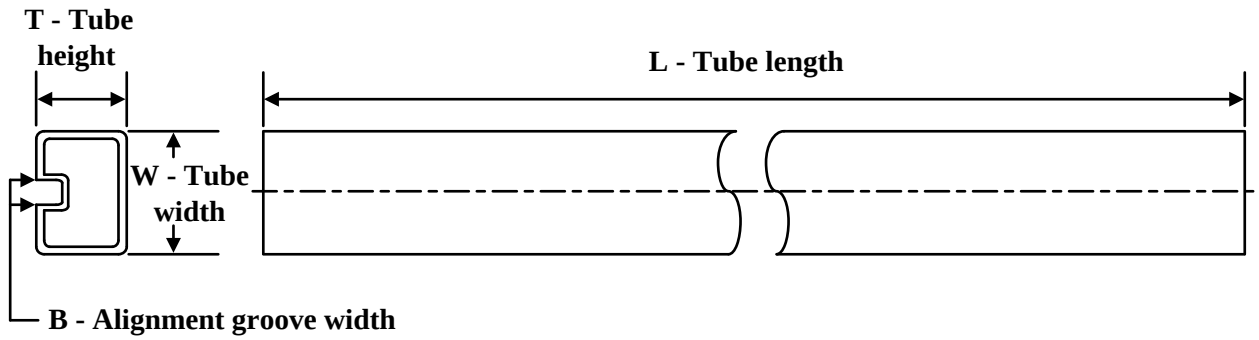
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
74LVCH16543ADLRG4	SSOP	DL	56	1000	356.0	356.0	53.0
SN74LVCH16543ADGGR	TSSOP	DGG	56	2000	356.0	356.0	45.0
SN74LVCH16543ADLR	SSOP	DL	56	1000	356.0	356.0	53.0

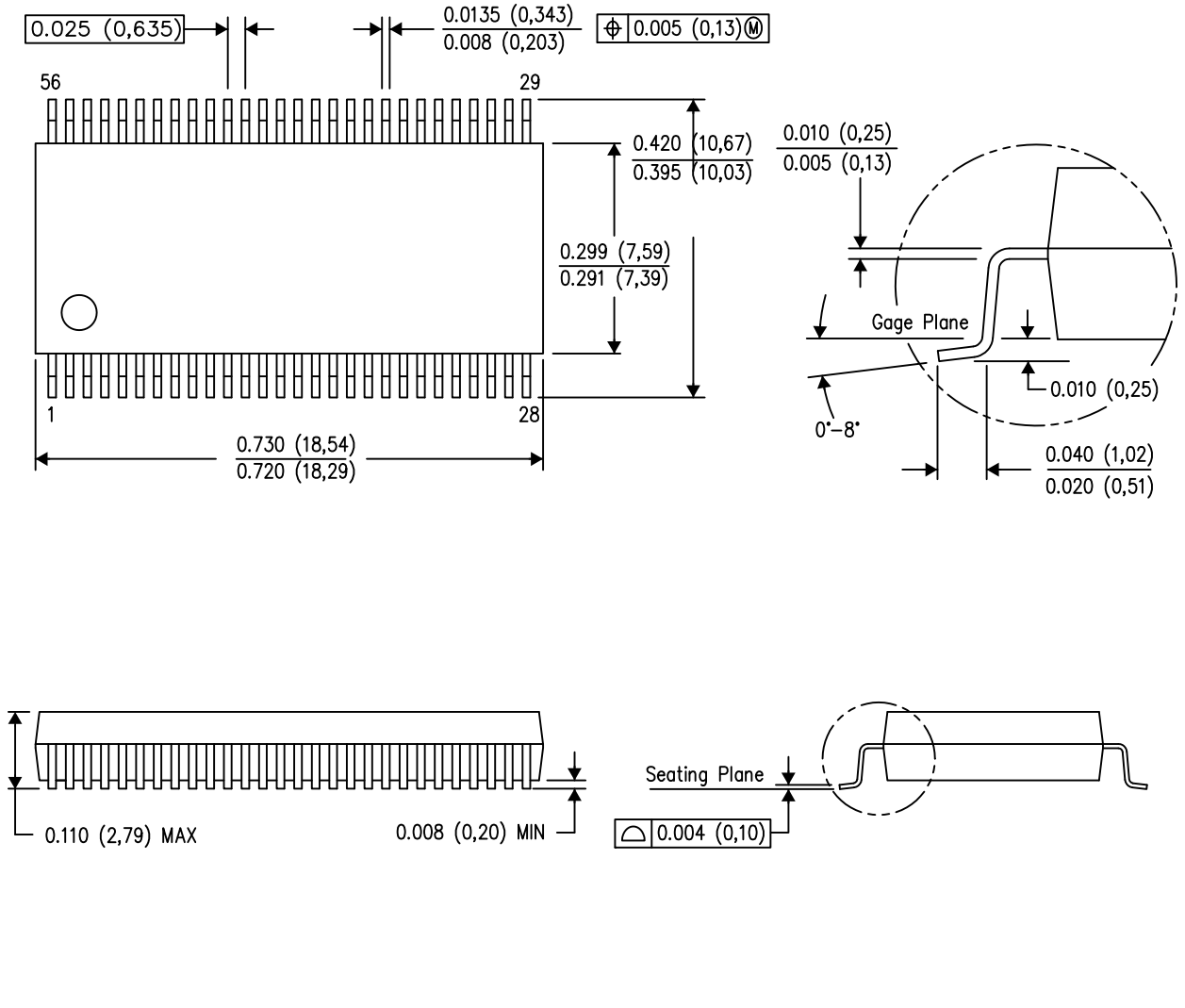
TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
SN74LVCH16543ADL	DL	SSOP	56	20	473.7	14.24	5110	7.87
SN74LVCH16543ADL.B	DL	SSOP	56	20	473.7	14.24	5110	7.87

DL (R–PDSO–G56) PLASTIC SMALL–OUTLINE PACKAGE



NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion not to exceed 0.006 (0,15).
D. Falls within JEDEC MO–118

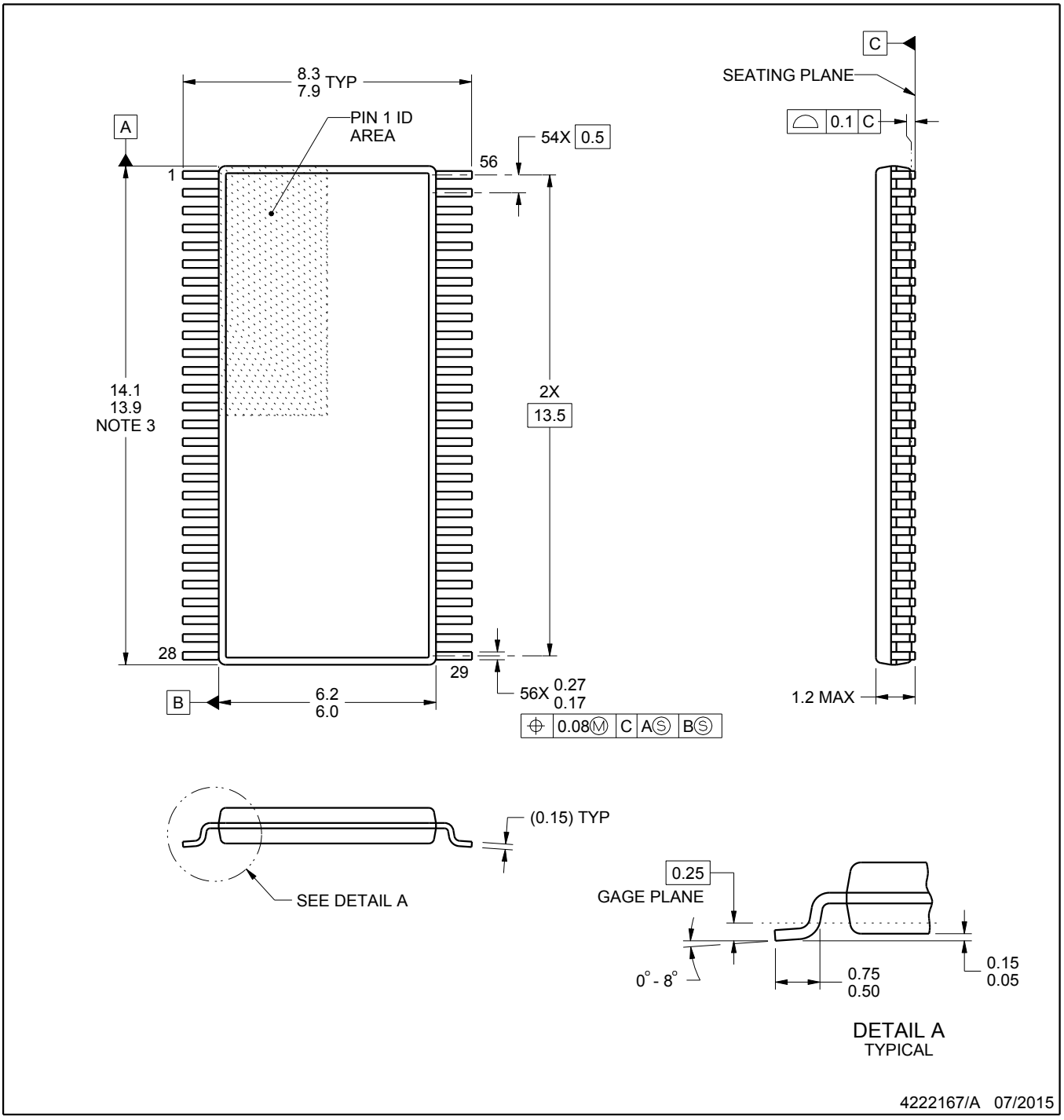
PowerPAD is a trademark of Texas Instruments.

PACKAGE OUTLINE

DGG0056A

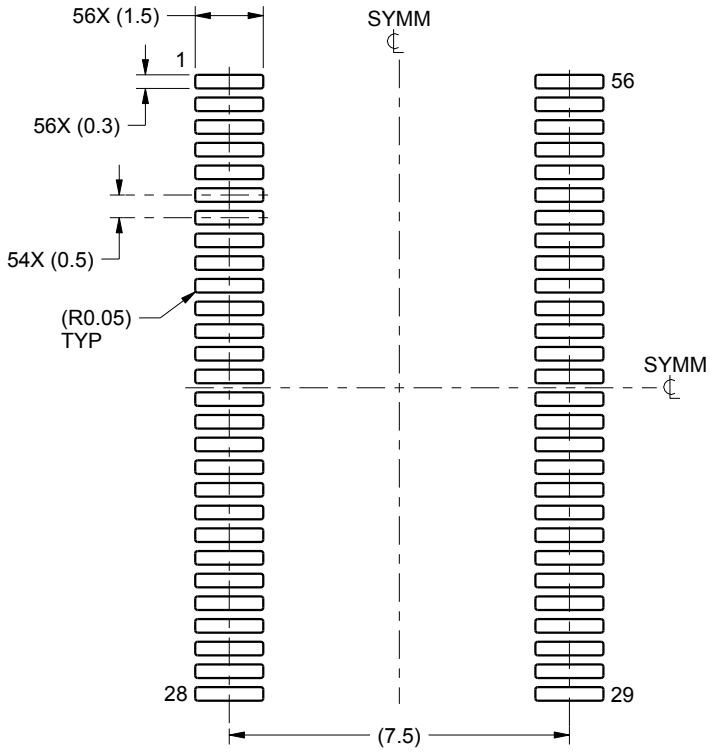
TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE

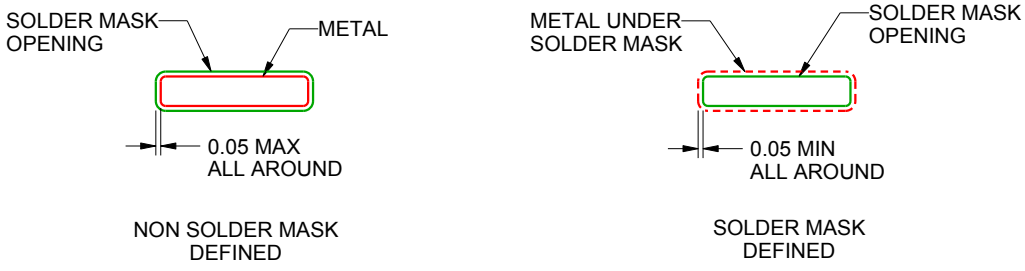


NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-153.



LAND PATTERN EXAMPLE
SCALE:6X

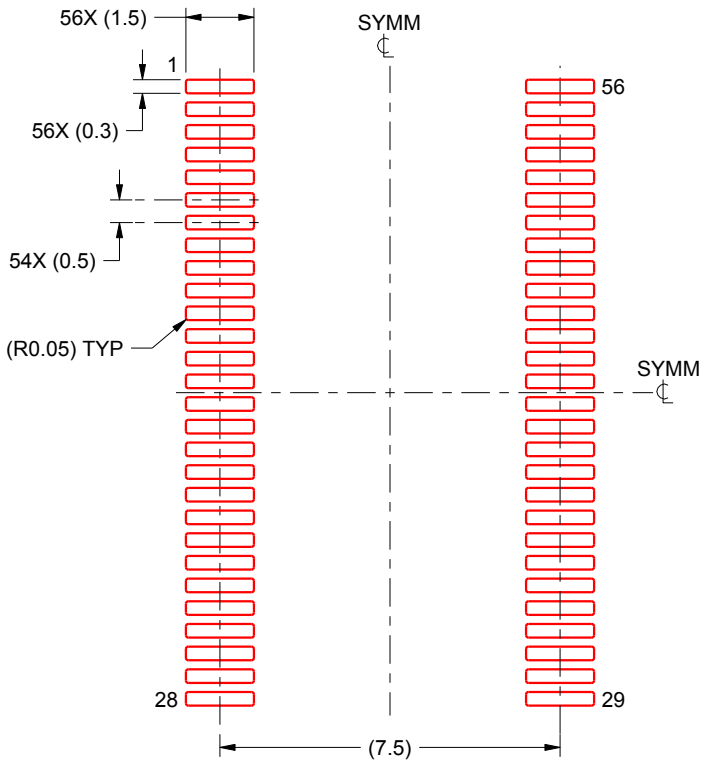


SOLDER MASK DETAILS

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NOTES: (continued)

5. Publication IPC-7351 may have alternate designs.
6. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:6X

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NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

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Last updated 10/2025